ASIS ANUPAM MOHANTY

aashishanup15 | Asis Anupam Mohanty | aashishanupam3@gmail.com | +91 9938680364

Summary

Electronics and Electrical Engineering undergraduate with hands-on industrial exposure in **PLC-based automation**, **SCADA** process control, and plant instrumentation, gained through training at NALCO Smelter Plant and CTTC Bhubaneswar. Proficient in **Siemens TIA Portal** and **Allen-Bradley RSLogix**, with working knowledge of embedded systems (Arduino/ESP32) and Python-based data analysis.

Work Experience & Industrial Training

Vocational Trainee (NALCO Smelter Plant, Angul) May 2026 – June 2026

- Completed one-month industrial vocational training in the **E&I C.R.L. & Telecommunication Department** at NALCO's integrated Smelter Plant.
- Gained practical exposure to industrial instrumentation including pressure transmitters, RTDs, thermocouples, LVDTs, flow/level switches, and load cells.
- Studied PLC hardware architecture, industrial automation systems, and SCADA-based process control used in aluminium smelting operations.
- Learned the working principles of analytical instruments including **XRF** and **OES** systems for material characterization and quality control.

PLC Training Trainee (CTTC, Bhubaneswar) May 2025 – June 2025

- Completed intensive hands-on training in industrial automation focusing on leading PLC platforms.
- Mastered **Ladder Logic (LAD)**, **FBD**, and **Structured Text (STL)** programming for industrial control applications.
- Gained practical experience with **Siemens TIA Portal V17** and **Allen-Bradley RSLogix** software environments.
- Applied principles of automation system design to real-time control scenarios and process troubleshooting.

Projects

Distance Measurement Device (Arduino)

- Built an ultrasonic sensor-based measurement unit with real-time LCD display.
- Implemented data averaging and filtering techniques to reduce sensor noise and improve accuracy.

Android Gamepad (ESP32)

- Designed a wireless Bluetooth game controller interfacing with Android devices.
- Optimized button mapping and serial communication for low latency performance.
- Analyzed input response behavior and improved system efficiency through iterative testing.

Autonomous EV Perception & Emergency Response System

– Designed a real-time perception system integrating object detection and acoustic signal analysis for autonomous vehicles.

– Implemented object detection using computer vision techniques with OpenCV for real-time obstacle identification.

– Developed an acoustic fusion module to detect emergency sirens and trigger automated response mechanisms.

Devanagari Script Recognition System

– Developed a machine learning-based system for recognizing handwritten and printed Devanagari characters using Python (NumPy, Pandas, OpenCV).

Education

2023 – 2027	**B.Tech in Electronics and Electrical Engineering**, KIIT University, Bhubaneswar (CGPA: 6.94, as of 6th Semester)	
2022	**Class 12th**, DAV Public School, CDA, Cuttack	(80.4%)
2020	**Class 10th**, DAV Public School, CDA, Cuttack	(95%)

Skills

Automation & Control	PLC Programming (LAD, FBD, STL), SCADA, Industrial Instrumentation, Siemens TIA Portal V17, Allen-Bradley RSLogix
Hardware Platforms	Arduino, ESP32, PLCs (Siemens, Allen-Bradley)
Core Competencies	Embedded Systems, Control Systems, Microcontroller Interfacing, Optimization Concepts
Software & Tools	Simulink, MATLAB, Jupyter Notebook
Programming & Data Tools	Python, C
Libraries & Frameworks	NumPy, Pandas, OpenCV (basic)
Machine Learning & Data	Machine Learning (Basics), Data Analysis, Data Preprocessing